SECOND BRAIN LABORATORIES · KNOWLEDGE + INTELLIGENCE

Regulatory Intelligence

The Financial Crisis and the Regulatory Maze

Institutional knowledge, typed relationships and evidence-grounded judgment

Laboratory 01 of four: a 10-cell pedagogical experiment linking an Obsidian knowledge vault, a regulatory graph, contextual retrieval, Claude-assisted analysis and an auditable export.

Alejandro Reynoso · October 2026

The central proposition: knowledge is not judgment

Knowledge asks

What does the institution know, how is it connected, and which source controls?

Intelligence asks

What matters for this crisis, which intervention is defensible, and what remains uncertain?

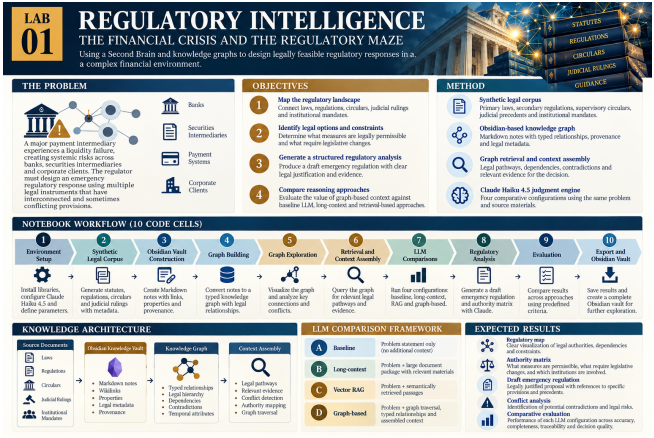
A Second Brain is a governed evidence architecture, not a larger prompt.

Laboratory 01: problem, objectives and method

The regulatory maze

- A payment intermediary suffers a systemic liquidity disruption.
- The regulator must identify powers, constraints and coordination requirements.
- Ten code cells connect knowledge representation to defensible analysis.

All legal texts in the laboratory are synthetic educational material.



The four laboratories: knowledge + intelligence

The common architecture

- Create an inspectable institutional memory.
- Represent relationships with explicit meanings.
- Assemble context for a concrete question.
- Apply bounded model judgment.
- Preserve evidence, uncertainty and an audit trail.

Why start with regulation?

- Different sources have different authority.
- A relevant text may not authorize action.
- Critical restrictions may be distant from the query.
- Decisions must be reviewable and time-sensitive.
- The same architecture transfers across domains.

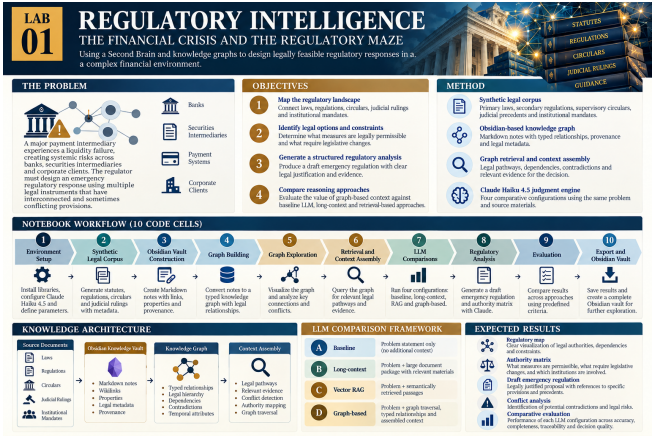
Laboratory 01 establishes the architecture; the course explores its transfer to further institutional settings.

Cell 01: establish a reproducible research environment

A governed starting point

- Install dependencies and set deterministic defaults for reproducible steps.
- Keep `ANTHROPIC_API_KEY` in Colab Secrets, never in the vault.
- Separate optional live LLM judgment from deterministic diagnostics.

The overview figure illustrates the laboratory as a whole; Cell 01 establishes its execution environment.

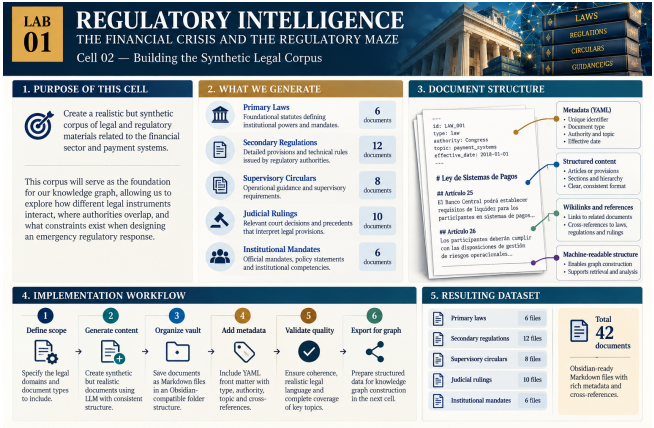


Cell 02: build a synthetic regulatory corpus

The institutional source layer

- Simulate primary laws, regulations, circulars, judicial rulings and mandates.
- Assign stable identifiers, types, source status and legal metadata.
- Enable controlled testing without claiming to state actual Mexican law.

Executable source of truth: 9 legal/interpretive documents within 16 graph objects; figures may portray earlier iterations.

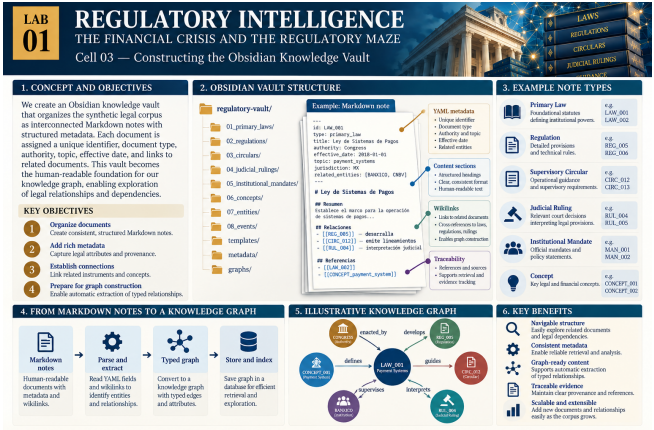


Cell 03: materialize the Obsidian knowledge vault

Knowledge becomes inspectable

- Export Markdown notes with structured YAML frontmatter.
- Use Obsidian wikilinks to expose source relationships.
- Preserve human readability alongside machine-processable metadata.

A note link alone does not identify the legal effect of its relationship.

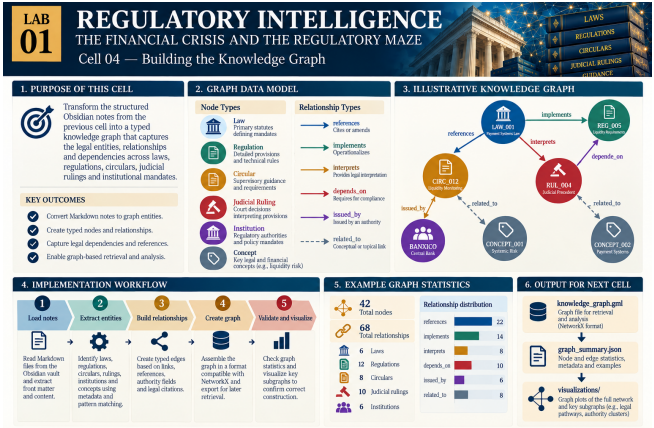


Cell 04: build the typed knowledge graph

Relations have meaning

- Create explicit nodes for sources, institutions, events and interventions.
- Distinguish authority, implementation, constraints and interpretation.
- Validate node and edge integrity before graph traversal.

Current executable graph: 16 objects and 26 typed edges.

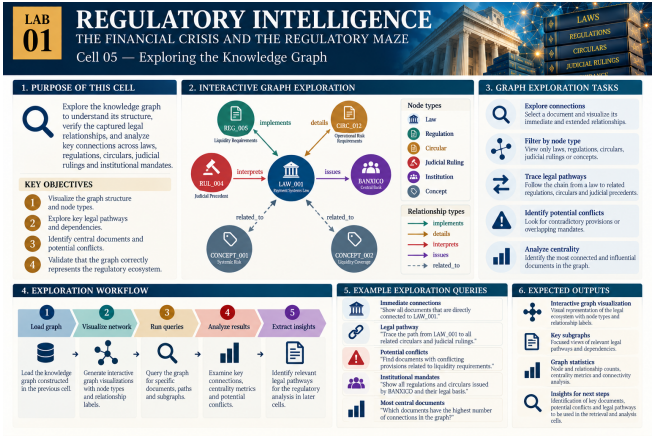


Cell 05: explore institutional connections

From files to legal pathways

- Explore neighbors, source types and institutional dependencies.
- Distinguish legal authority from mere topical proximity.
- Examine possible conflicts and central nodes.

The annotated visual describes graph exploration; the monograph frames the executable Cell 05 around a crisis decision mandate.

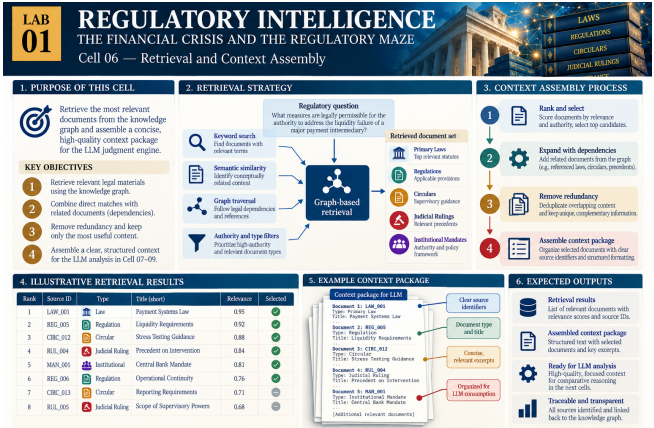


Cell 06: retrieve and assemble context

The institutional relevance problem

- Use the shock and proposed interventions as decision anchors.
- Retrieve relevant documents and follow legal dependencies.
- Package concise evidence with source IDs and rationale.

The current notebook also establishes controlled retrieval comparison baselines at this stage.

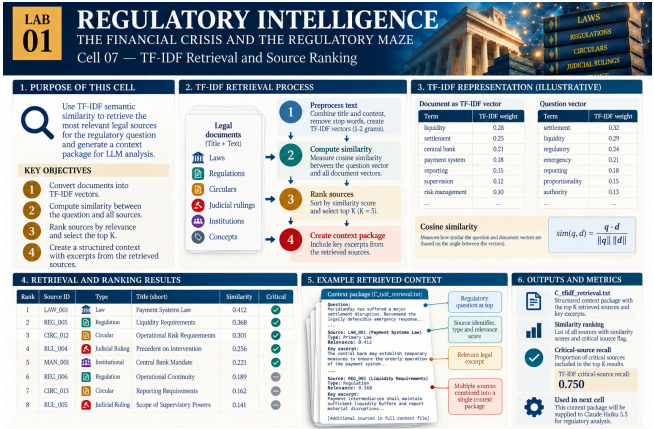


Cell 07: rank evidence with TF–IDF

A transparent baseline

- Represent source texts and the question as weighted lexical vectors.
- Rank documents by cosine similarity and select the top candidates.
- Evaluate whether lexical relevance recovers critical authority.

Top-5 TF–IDF recovers five of six critical evidence IDs in the current experiment.



Cell 08: assemble graph-aware context

Add structural explanations

- Traverse relationships from the crisis to potentially decisive sources.
- Retain pathways that explain why a document was included.
- Send the assembled package to the bounded judgment layer.

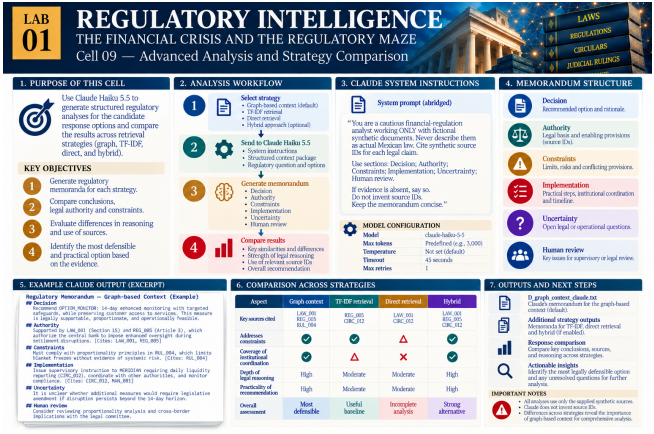
The supplied graphic reflects an earlier cell label: in the current executable notebook, Cell 08 builds graph context, not the final model memorandum.

Cell 09: constrain the LLM judgment

Judgment downstream of evidence

- Ask Claude to compare interventions only against supplied synthetic documents.
- Require Decision, Authority, Constraints, Implementation, Uncertainty and Human review.
- Cite note IDs; never invent statutes or fabricate a model response.

Live LLM results are distinct from deterministic evidence diagnostics.

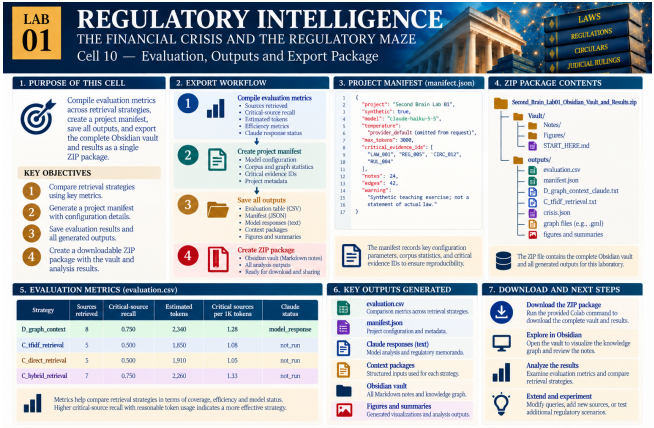


Cell 10: evaluate, document and export

An auditable institutional artifact

- Record recall, source count, token estimates and model status.
- Preserve the manifest, context packages, outputs and knowledge vault.
- Export a portable ZIP to support independent review and re-execution.

The final artifact is an evidence package, not simply a fluent answer.



What the controlled comparison demonstrates

Context strategy	Sources	Recall	Est. tokens	Critical IDs
A: Problem only	0	0.000	88	0/6
B: Long context	9	1.000	568	6/6
C: TF–IDF retrieval	5	0.833	366	5/6
D: Graph context	9	1.000	672	6/6

The strongest observation

Long context and graph context both expose all six critical sources on this small corpus.

The graph's contribution

Structural paths explain *why* an authority or restriction entered the context.

These are notebook-reported, deterministic evidence metrics, not proof of legal accuracy or guaranteed LLM superiority.

Institutional knowledge is a governed asset

Knowledge governance

- Authenticate source identity and provenance.
- Track versions, effective dates and amendments.
- Review claims about relationships and authority.
- Preserve contested interpretations rather than overwriting them.

Judgment governance

- Separate evidence selection from model inference.
- Validate citations and unsupported conclusions.
- Surface uncertainty and constraints.
- Require accountable institutional review before action.

The evidence path must remain inspectable even when the original analyst is absent.

Beyond the prototype: the production challenge

What must be added

- Verified primary legal sources.
- Effective-time and jurisdiction filters.
- Relation-aware graph traversal.
- Edge provenance, reviewer roles and source hashes.
- Reliable evaluation of legal reasoning quality.

The decisive classroom experiment

- Change a synthetic legal constraint.
- Rebuild the vault and graph.
- Recalculate retrieval and context packages.
- Examine whether the recommendation changes.
- Audit exactly where and why it changed.

The true test is not the first answer; it is whether changed institutional knowledge changes judgment in an intelligible way.

Further reading: intellectual foundations

- Bush (1945), "As We May Think."
- Engelbart (1962), *Augmenting Human Intellect*.
- Simon (1947), *Administrative Behavior*.
- March and Simon (1958), *Organizations*.
- Nonaka and Takeuchi (1995), *The Knowledge-Creating Company*.
- Davenport and Prusak (1998), *Working Knowledge*.
- Alavi and Leidner (2001), knowledge-management systems.
- Weick (1995), *Sensemaking in Organizations*.
- Berners-Lee, Hendler and Lassila (2001), semantic web.
- Angles and Gutierrez (2008), graph database models.
- Hogan et al. (2021), *Knowledge Graphs*.
- Lewis et al. (2020), retrieval-augmented generation.
- Liu et al. (2024), "Lost in the Middle."
- Additional source-specific references and methodological qualifications appear in the full monograph.

Presentation reading selections, condensed from the monograph bibliography.

LABORATORY 01 · CONCLUSIONS

From knowledge to governed judgment

The Second Brain does not merely remember sources.
It reconstructs why evidence matters for a decision.

Institutional intelligence combines governed knowledge, bounded judgment and accountable human review.

Alejandro Reynoso · October 2026